

Machine Learning 2 26
(CSE 343 / ECE 363)
Assignment 1

Max Marks: 90 **Due Date:** Theoretical - 17 September 2026,
Practical - 07 October 2026

Instructions

Plagiarism Policy: The assignment must be attempted individually. Collaboration is allowed only for high-level discussions. **Copying, plagiarism, or sharing of solutions/code will be dealt with strictly.**

Submission: Submit a **single ZIP** file named:

`A1_Roll_Number.zip`

For example: `A1_2024000.zip`.

The ZIP must contain exactly:

- `Roll_Number_Theory_a1.pdf` — handwritten theory solutions.
- `Roll_Number_Practical_a1_q3.ipynb` — Question 2(d) and 3 notebook.
- `Roll_Number_Practical_a1_q4.ipynb` — Question 4's regression notebook.

Theory: All theory questions must be **handwritten** and submitted in the theory PDF.

Practical: Use **Python only**. The notebook must contain all required code, preprocessing, analysis, results, and visualizations. Document observations and answers in Markdown cells.

Reproducibility: Use your **roll number as the random seed** wherever applicable. For roll numbers containing letters, use their ASCII values as specified by the course instructions. **Failure to set the required seed may result in a penalty.**

Submission Policy: Upload the ZIP on Google Classroom and click **Turn In**. Late submissions will incur penalty as per the course policy. In case of multiple submissions, the latest submission before the deadline will be considered.

Checklist: Ensure the ZIP contains correctly named files, all required code/results, readable handwritten solutions, required visualizations, and reproducible results.

Q1. [10 points] Grouped (Binomial) vs. Individual (Bernoulli) logistic regression [Theory]

The lecture notes that when multiple individuals share the same x (e.g., same age group), y can be modeled as $\text{Binomial}(n, p)$ rather than $\text{Bernoulli}(p)$ per individual.

- (a) **[4 points]** Write the log-likelihood $\ell(\cdot)$ for a dataset of M groups, where group k has n_k trials and y_k successes, using the same logit link $p_k = \sigma(w^\top \mathbf{x}_k)$.
- (b) **[6 points]** Prove that maximizing this grouped log-likelihood gives the *exact same* $\hat{w}$ as maximizing the ungrouped, per-individual Bernoulli log-likelihood over the same raw data. What does this tell you about when grouping data changes your estimate and when it doesn't?

Q2. [18 points] Parameter Estimation in Logistic Regression [Theory & Practical]

Given a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ where the inputs are vectors $x_i \in \mathbb{R}^d$ and the binary targets are $y_i \in \{0, 1\}$, a binary logistic regression model estimates the posterior class probabilities using the sigmoid activation function $\sigma(z) = \frac{1}{1 + e^{-z}}$:

$$P(y_i = 1 \mid x_i, w) = \sigma(w^\top x_i) = \frac{1}{1 + e^{-w^\top x_i}}$$

- (a) **[4 points]** Assuming the data points are independent and identically distributed (i.i.d.), write down the joint conditional likelihood function $P(Y \mid X, w)$. Show that maximizing this joint conditional likelihood with respect to w is mathematically equivalent to minimizing the empirical binary cross-entropy loss function $J_{\text{MLE}}(w)$.
- (b) **[5 points]** Place an independent zero-mean Gaussian prior over the weight vector $w \sim \mathcal{N}(0, \sigma_0^2 I)$. Prove that maximizing the log-posterior $\log P(w \mid X, Y)$ corresponds to minimizing:

$$J_{\text{MAP}}(w) = J_{\text{MLE}}(w) + \lambda \|w\|_2^2$$

Express the regularizer weight λ explicitly in terms of the prior variance σ_0^2 .

- (c) **[3 points]** Explain mathematically why the MAP estimate $\hat{w}_{\text{MAP}}$ approaches the MLE estimate $\hat{w}_{\text{MLE}}$ as sample size $N \rightarrow \infty$.
- (d) **[6 points]** Write a Python script using scikit-learn and matplotlib to evaluate the asymptotic convergence of Logistic Regression parameters:
 - (i) **[2 points]** Generate a synthetic binary classification dataset using true weight vector $w^* = [1.5, -2.0, 0.5]^\top$. Evaluate sample sizes:

$$N \in \{15, 25, 50, 100, 200, 400, 800, 1600, 3200, 6400, 12800\}$$

- (ii) **[2 points]** For each dataset size N : Fit Unregularized MLE and MAP with Gaussian Prior.

- (iii) [2 points] Plot $w_{1 \text{ MLE}}$, $w_{1 \text{ MAP}}$, and true parameter line $w_1^* = 1.5$ vs N on a semilog-x scale.

Q3. [22 points] Classification & Diagnostic Analysis with Logistic Regression
[Practical]

Target Dataset: Scikit-learn's Breast Cancer Diagnostic Dataset

1. Exploratory Data Analysis & Preprocessing

1.1 [2 points] Class Distribution Check:

Compute the raw counts and percentages for each target class (0 = Malignant, 1 = Benign).

Based on these percentages, is this dataset balanced or imbalanced?

1.2 [3 points] Multicollinearity Analysis:

Plot a correlation heatmap for the first 10 numerical features in the dataset.

Identify at least two feature pairs that exhibit high collinearity ($r > 0.85$).

Why does severe multicollinearity pose a structural problem when interpreting logistic regression coefficients?

1.3 [2 points] Train-Test Split & Feature Scaling:

Perform an 80/20 train-test split.

Apply `StandardScaler` to scale the feature matrix.

Why must `fit_transform()` be applied strictly to the training data, while only `transform()` is applied to the test data?

2. Model Training & Feature Coefficient Analysis

2.1 [1 points] Baseline Model Training:

Fit a `LogisticRegression` baseline model on the scaled training dataset.

2.2 [6 points] Odds Ratio Extraction:

Extract the model coefficients () and calculate the corresponding Odds Ratios ($= \exp()$).

Construct a pandas `DataFrame` mapping feature names to their coefficients and odds ratios, sorted by coefficient magnitude.

Generate a horizontal bar chart displaying the 5 most positive and 5 most negative coefficients.

Select one feature with a strong negative coefficient. Explain how a 1 standard-deviation increase in this feature alters the odds of the tumor being class 1 (Benign).

What structural role does L_2 regularization play on coefficient magnitudes?

3. Diagnostics, and Threshold Optimization

3.1 [3 points] Default Threshold Evaluation ($p = 0.5$):

Generate class predictions on the test set.

Display the resulting Confusion Matrix and Classification report.

In a medical diagnostic pipeline designed to detect cancer, which error type is more dangerous: False Positives or False Negatives? Justify your reasoning.

3.2 [5 points] Custom Decision Threshold Optimization:

Write a function to iterate through decision thresholds from 0.10 to 0.90 in increments of 0.05.

Identify an optimal decision threshold that satisfies a minimum Recall of 95% for the Malignant class while maximizing overall Precision.

Display the updated Confusion Matrix under this custom threshold.

Q4. [40 points] Regression: End-to-End Pipeline [Practical]

Submission note for this part Rename the notebook with your roll number before submitting (e.g. roll number 2024000 $\rightarrow$ 2024000_q3.ipynb). Extract Q3_DATA.zip next to the notebook. Write code only in sections marked `YOUR CODE HERE` — you can NOT write code anywhere else, and you may not modify any function or class definition outside such a region, nor add or delete cells. Seed every random operation with your roll number. Do not import any library beyond what is already provided; every part is solvable without one. Your notebook is re-run after the deadline on data it has not seen — if it fails to run top to bottom, no compensation will be given. Notebook parts are graded by re-running your code; report parts are graded by hand, and are where you justify the choices you made in the notebook.

- (a) [1 point] **Data Loading.** Load the raw training data so it is ready to explore.
- (b) [4 points] **Exploratory Data Analysis.** Explore the raw data. Look at what kind of features you have, missing values, the target. Put your EDA analysis plots and tables that mattered in your report. (You would have to justify your choices.)
- (c) [8 points] **Data Preparation and Feature Engineering.**
 - (i) [4 points] Based on the observations from your exploratory data analysis, prepare the dataset for modeling. Clearly explain the preprocessing and feature-engineering decisions you make and how they are motivated by patterns or

issues identified during EDA. Only short bullet points are needed in the report, not long explanations.

- (ii) **[4 points]** Evaluate whether your chosen data preparation and engineered features are appropriate for modelling. Provide evidence for your choices and discuss how you ensured that the resulting pipeline remains valid when applied to previously unseen data.

A submission that fails part (i) or (ii) also scores zero on the performance parts of parts (e) and (g) below.

- (d) **[upto 3 points] Validation Strategy.** Split the training data into a training part and a validation part, and state your strategy briefly for the same. Justify why that split is sound for training.
- (e) **[6 points] Model Training and Evaluation.** Your score here is computed twice: once in this part, and again in part (g) after you iterate on it. The two are graded separately.
 - (i) **[up to 4 points]** Train a linear regression model on your split. Scored on a hidden test set:

R ² on hidden test set	Marks
≥ 0.90	4
0.85 to 0.90	3
0.75 to 0.85	2
0.65 to 0.75	1
below 0.65	0

- (ii) **[2 points]** Justify your choice of model and hyperparameters. Before looking at part (f)'s diagnostics, say what you expect this choice to do to the train and validation gap.
- (f) **[6 points] Diagnostics.**
 - (i) **[2 points]** Produce diagnostics for your trained model: the gap between train and validation performance, a learning curve for both, and the rows it gets most wrong.
 - (ii) **[2 points]** Look at the train and validation gap and the learning curve. Say what they tell you about your model.
 - (iii) **[2 points]** Look at the rows your model gets most wrong. What is responsible for this behavior (you should be able to justify it by EDA analysis).
- (g) **[9 points] Model Improvement.**
 - (i) **[up to 4 points]** Take what you learned from part (f) and change something: your model, its hyperparameters, or how you are using your features. Retrain. Scored on the same hidden test set as part (e):

R on hidden test set	Marks
≥ 0.90	4
0.85 to 0.90	3
0.75 to 0.85	2
0.65 to 0.75	1
below 0.65	0

- (ii) **[3 points]** Explain what you changed and why, based on your diagnostics.
- (iii) **[2 points]** Compare this score with part (e)'s, and say whether the change helped.
- (h) **[3 points] Generalization and Limitations.**
 - (i) **[1 point]** The hidden test set contains groups you have never seen. Give one concrete way your pipeline could still fail on them even with a good score.
 - (ii) **[2 points]** Name one change you would try next, to the data, the features, the model, or the validation strategy, and explain why you expect it to help the most.